Diffusion: from training pairs to generation

ES 667 · Generative Modelling II · Two-page recap

1. Define the corruption process

x_0 is clean data; x_t is a noisy state. Choose variance β_t, set $\alpha_t = 1 - \beta_t$, and cumulative retention $\bar{\alpha}_t = \prod_{s=1\ldots t} \alpha_s$, with $\bar{\alpha}_0 = 1$.

One transition: $x_t = \sqrt{\alpha_t}\, x_{t-1} + \sqrt{\beta_t}\, \eta_t$, with independent $\eta_t \sim N(0,I)$.

Direct marginal: $x_t = \sqrt{\bar{\alpha}_t}\, x_0 + \sqrt{(1-\bar{\alpha}_t)}\, \varepsilon$. Sharing ε across times gives correct marginals, but a different joint coupling from a recursive trajectory.

2. Manufacture a supervised target

Sample x_0 from the dataset, integer t uniformly from 1…T, and independent $\varepsilon \sim N(0,I)$. Construct x_t. Feed only (x_t, t) to the unconditional denoiser; compare its prediction with ε. Average squared errors over examples and coordinates. Backpropagate through the denoiser and update θ.

Population MSE estimates $E[\varepsilon \mid x_t, t]$, not each hidden random draw. The network does not receive the clean image or target noise as extra inputs.

3. Keep three values distinct

Noisy state x_t; estimated clean image $\hat{x}_0$; sampled previous state x_{t-1}. The estimate $\hat{x}_0 = (x_t - \sqrt{(1-\bar{\alpha}_t)}\hat{\varepsilon})/\sqrt{\bar{\alpha}_t}$ is not the next chain state.

Scalar fixture: $\beta_1 = \beta_2 = 0.2$, $x_0 = 1$, $\varepsilon = -0.5 \rightarrow x_2 = 0.5$. For $\hat{\varepsilon} = -0.4$: loss=0.01, $\hat{x}_0 = 0.925$, reverse mean=0.708088, variance=1/9, standard deviation=1/3. Fresh $z = 0.3$ gives $x_1 = 0.808088$.

4. Remember the ELBO qualification

The negative variational bound contains a terminal prior KL, expected local reverse KLs, and a negative endpoint log likelihood. For t>1, fixed reverse covariance gives a weighted noise error with $w_t = \beta_t^2/[2\sigma_t^2\alpha_t(1-\bar{\alpha}_t)]$. Unweighted MSE changes that weighting; it is not the entire exact negative ELBO.

Source: Ho, Jain & Abbeel, DDPM (2020)

Generation, conditioning and evidence

5. Execute the sampler

Start from $x_T \sim N(0,I)$. For t=T…1, predict $\hat{\varepsilon}=\varepsilon_\theta(x_t,t)$. Compute $\mu=[x_t-\beta_t\hat{\varepsilon}/\sqrt{(1-\bar{\alpha}_t)}]/\sqrt{\alpha_t}$. Set $\sigma^2=\beta_t(1-\bar{\alpha}_{t-1})/(1-\bar{\alpha}_t)$. For t>1 draw $x_{t-1}=\mu+\sigma z$; at t=1 return μ without new noise. Model parameters stay fixed.

This recurrence uses one-step α in the denominator. Skipping arbitrary steps is not a defined faster sampler. The teaching model uses 1,000 linear-schedule steps and terminal $\bar{\alpha} \approx 0.00004036$.

6. Assemble text-to-image components

A spatial autoencoder may map [B,3,512,512] to [B,4,64,64]: 64× fewer spatial positions but 48× fewer scalars. Those are representation ratios, not guaranteed runtime gains. Train the autoencoder, freeze it in the standard example, and train diffusion on correctly scaled latents. At ordinary text-to-image inference: latent noise → repeated denoising → decode once.

In the selected cross-attention block, image features supply Q; text features supply K,V. $A=\mathrm{softmax}_{\text{text positions}}(QK^T/\sqrt{d_k})$; AV retains one row per image query. Original DiT is class-conditioned; different systems use different conditioning designs.

Classifier-free guidance: $\varepsilon_{cfg}=\varepsilon_u+s(\varepsilon_c-\varepsilon_u)$. s=0 is unconditional; s=1 is ordinary conditional; s>1 extrapolates. Conditional dropout teaches both predictions using shared weights. Define the null condition explicitly.

7. Change the state space for masked language

Token IDs are arbitrary lookup indices. A discrete mask process replaces positions with [MASK]; the model predicts vocabulary logits [B,L,V]. Simple masked-position CE is pedagogical. The stated LLaDA-style uniformly sampled-time objective adds a 1/t weight. Specify response-slot count, observed prompt tokens, reveal/remask policy and sampling schedule. Parallel logits do not alone establish faster generation.

8. Read the evidence correctly

The new MNIST run: 814,145 parameters, 2,000 steps, seed 667, batch 64. Its unselected samples show malformed and similar digits. The older guidance example uses a separate 4,000-step class-conditioned cosine-schedule model. The eight-cluster demo is analytical; text traces and attention features are illustrative. A plausible scientific reconstruction is not a measured observation. No FID or downstream gain is asserted.

Latent diffusion · Classifier-free guidance · LLaDA